# Fish, amphibians, and reptiles

THESE GROUPS ARE THE MOST PRIMITIVE VERTEBRATES (ANIMALS WITH BACKBONES).

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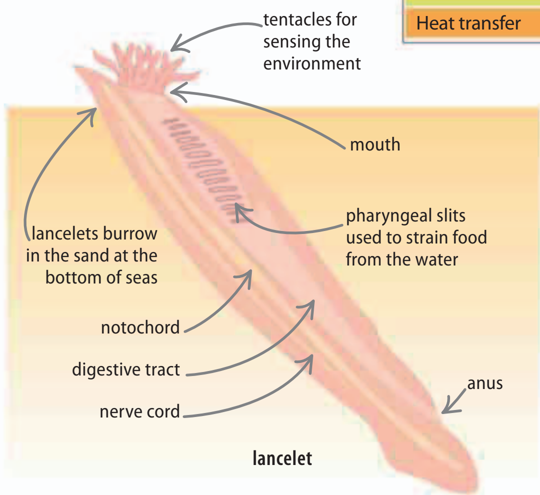
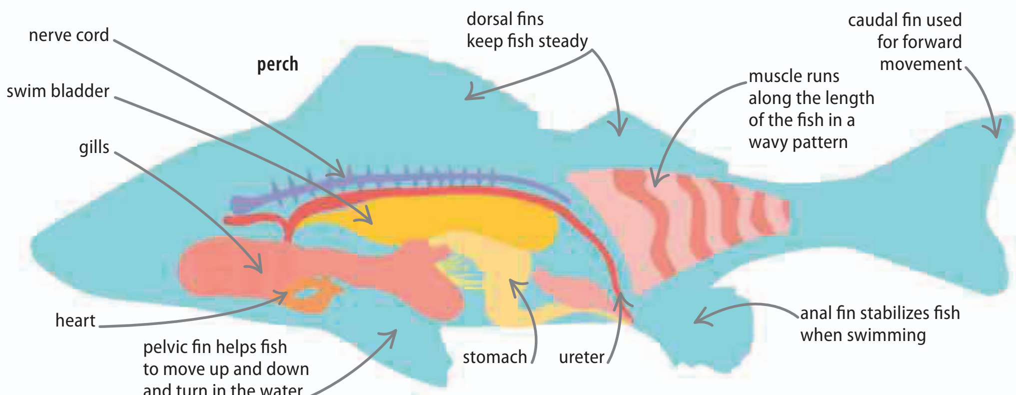
**Fish, amphibians, and reptiles are three classes of vertebrates, the group to which birds and mammals—including humans—belong.**

## What is a vertebrate?

Vertebrates make up most of the phylum Chordata. "Chordata" refers to a flexible supporting rod, called the notochord, that is present at some point in the life of all chordates. In most cases, it develops into a vertebral column—a chain of interlinked bones that form the spine, or backbone. This protects a spinal cord, a thick nerve bundle that connects the brain to the rest of the body.



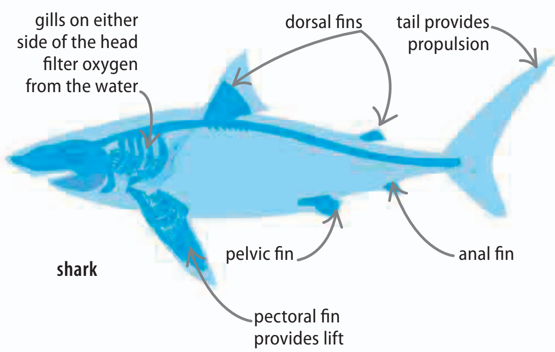

◁ **No skull**
The first vertebrates are thought to have looked like today's lancelets, simple aquatic animals that live on the seabed. Lancelets have no skull, unlike true vertebrates, but they share other features, including a notochord and pharyngeal slits (which form gills in fish).




◁ **Bony fish**
Bony fish, unlike cartilaginous fish, can control their buoyancy by altering the levels of gas in an internal float called the swim bladder.

## Fish

Several groups of fish have risen and fallen since the first fish evolved 500 million years ago (mya). There are two main groups of fish living in the world's marine and freshwater habitats today. The first have skeletons of bone and number about 20,000 species. The other group of 800 species includes the sharks and rays, which have skeletons made from cartilage—the same flexible tissue found in the outer part of the human ear.




◁ **Cartilaginous fish**
A shark's cartilaginous skeleton (in dark blue) and streamlined body shape help it move quickly through water. Flexible rods of cartilage stiffen the flat fins and tail lobes. The dorsal fins keep the shark from rolling over as swishes of its long tail power it through the water.